Saturated Superfluid Vacuum VII-a

Quantum Mechanics from Hydrodynamics

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Draft

Abstract

This paper derives the quantum-mechanical limit of the Saturated Superfluid Vacuum (SSV) framework. The scope is deliberately narrow: the Madelung transformation, the linear Schrödinger limit, uncertainty as a hydraulic localisation bound, and measurement as a physical reconnection channel.

The formal result is the recovery of the Schrödinger equation from the low-amplitude hydrodynamic regime of the SSV medium. Quantisation enters as the single-valuedness condition on the order-parameter phase, tying discrete winding numbers to circulation topology rather than to a separate quantisation rule. Uncertainty is interpreted as the hydraulic cost of localising the order parameter. Measurement is modelled as an irreversible reconnection or wake-writing event in the medium.

Claims about emergent geometry, Einstein equations, cosmogony, and full-series synthesis are left to companion papers.

Contents

1 Introduction

The SSV programme treats the vacuum as a saturated superfluid described by a complex order parameter Ψ. Papers I–IV establish the medium, its topological defects, thermodynamic arrow, and gravitational limit. Paper V extends the same picture to strong gravity and condensate black holes. The present paper asks a narrower question: under what conditions does ordinary quantum mechanics appear as the low-amplitude hydrodynamic limit of this medium?

The working thesis is that quantum behaviour is not an additional foundational axiom. It is the wave-mechanical regime of the order parameter when perturbations are small, gradients are gentle, and defects can be treated through an effective potential. In that regime the Madelung variables map the complex wave equation to continuity plus Euler-type flow equations, and the inverse map recovers the Schrödinger equation for a slowly varying envelope.

This paper separates three claim levels explicitly.

- **C1. Formal derivation.** The Madelung transformation and its inverse show how a Schrödinger-type equation is equivalent to hydrodynamic flow with a quantum-pressure term. This is an exact algebraic equivalence, not an approximation.
- **C2. Controlled limit.** In the low-amplitude, long-wavelength regime, the nonlinear medium terms can be absorbed into background phase and effective potentials, leaving the ordinary linear Schrödinger equation. This is the limit that makes SSV contact with standard quantum mechanics.
- **C3. Physical interpretation.** Quantisation, uncertainty, and measurement are read as circulation topology, localisation stiffness, and irreversible reconnection respectively. These are model commitments that arise naturally from the medium picture; they are not independent mathematical theorems.

Roadmap

Section 2 carries out the Madelung transformation and derives the hydrodynamic equations. Section 3 takes the Schrödinger limit and shows how quantisation follows from phase topology. Section 4 interprets the uncertainty relation as a hydraulic localisation bound. Section 5 models measurement as a reconnection event and discusses Born weights. Section 6 places the results in the context of the series and lists open calculations.

2 Madelung Transformation

2.1 Polar decomposition

Let the local state of the saturated medium be the complex order parameter

$$\Psi(\mathbf{x},t) = \sqrt{\rho(\mathbf{x},t)}\, e^{iS(\mathbf{x},t)/\hbar}, \tag{1}$$

where ρ is the mass density and S is the phase action.

Normalisation convention, stated once (#189). ρ is used in two conventions in this paper and the difference is recorded here rather than left to the reader. In the hydrodynamic sections (4)–(7) and in the Schrödinger limit, $\rho = |\Psi|^2$ is a *mass density* and ρ_0 is the saturated background of Papers I–IV. In §4.1 the single-particle convention $\int |\Psi|^2\, dx = 1$ is used instead, without which (17) would not give $\sigma^2/2$; there $|\Psi|^2$ is a probability density and ρ_0 is whatever reference density makes the logarithm's argument dimensionless. The two differ by a factor of the particle mass and a normalisation volume. Nothing in (4)–(7) depends on the choice — Q

in (6) is invariant under $\rho \to \lambda\rho$ — but the symbol does carry two dimensions across the paper, and saying so is cheaper than the alternative. Generalised as #205.

The associated irrotational velocity field is

$$\mathbf{v} = \frac{1}{m}\nabla S. \tag{2}$$

For the logarithmic-Schrödinger (LogSE) medium of Papers I–IV [1, 2], the order parameter obeys

$$i\hbar\,\partial_t\Psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{ext}} + U(\rho)\right]\Psi, \tag{3}$$

where V_{ext} is a slowly varying effective potential and $U(\rho)$ is the medium self-interaction. The specific equation of state is fixed in earlier papers; the present paper uses only the hydrodynamic structure common to this class of order-parameter equations.

2.2 Hydrodynamic form

Substituting (1) into (3) and separating real and imaginary parts gives the Madelung pair

$$\partial_t\rho + \nabla\cdot(\rho\mathbf{v}) = 0, \tag{4}$$

$$\partial_t S + \frac{(\nabla S)^2}{2m} + V_{\text{ext}} + U(\rho) + Q = 0, \tag{5}$$

with the quantum pressure

$$Q(\mathbf{x},t) = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}. \tag{6}$$

Taking a gradient of (5) gives the Euler equation

$$m(\partial_t + \mathbf{v}\cdot\nabla)\mathbf{v} = -\nabla\big(V_{\text{ext}} + U(\rho) + Q\big). \tag{7}$$

Madelung equivalence

Equations (4)–(7) are the Madelung equations: compressible irrotational hydrodynamics with an additional quantum-pressure contribution Q. The transformation is exact and bidirectional: starting from the wave equation (3) yields the hydrodynamic pair; starting from the hydrodynamic pair with the irrotationality condition (2) reconstructs the wave equation.

3 Schrödinger Limit

3.1 Low-amplitude long-wavelength limit

Write the density as $\rho = \rho_0 + \delta\rho$, with $|\delta\rho| \ll \rho_0$, and assume gradients are long compared with the healing length of the medium. Expand the self-interaction around the saturated background:

$$U(\rho) = U(\rho_0) + U'(\rho_0)\,\delta\rho + \cdots \tag{8}$$

The constant term shifts the global phase; the linear term is either negligible for a sufficiently weak perturbation or is absorbed into an effective potential. The slowly varying envelope ψ then obeys

$$i\hbar\,\partial_t\psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{eff}}\right]\psi, \tag{9}$$

the ordinary Schrödinger equation for the effective degree of freedom.

Schrödinger limit

The SSV medium in its low-amplitude, long-wavelength regime reduces exactly to linear Schrödinger dynamics. The claim is not that all of quantum theory is derived in one step; it is that whenever the conditions above hold, the same hydrodynamic variables reduce to the linear dynamics used in standard quantum mechanics.

Origin of V_{eff}, and the hydrogen spectrum. The effective potential in (9) is not imported from elsewhere; it is the linearised long-wavelength shear field around the source defect. For a heavy stationary source defect with topological chirality (Paper I, electric-charge identification), the chiral-shear sector of the SSV Lagrangian propagates the source's chirality outward at the transverse phase speed $c_\perp = \alpha c$. At distances large compared with the source healing length and small compared with the cosmological horizon, the linearised chiral-shear potential reduces to the Coulomb form

$$V_{\text{eff}}(r) = -\frac{e^2}{4\pi\varepsilon_0\, r}, \tag{10}$$

where the proportionality is fixed by the same identification $\alpha = c_\perp/c$ that defines the chiral-shear coupling in Paper I (§ Postulate).

The reduction to Coulomb form is Paper II's [3]: Paper I fixes $\alpha = c_\perp/c$ and routes the statics of charged defects to Paper II, where the Bernoulli-pressure derivation of $F_C = \alpha\hbar c/r^2$ is carried out.

What (10) costs should be said plainly. Paper II records the Coulomb coupling as *empirical* α: the $1/r$ form is derived, the strength is an input. So what follows is a *consistency* result — the Rydberg series is recovered given $\{m_e, \alpha, c\}$, which the series already concedes as inputs — and not a prediction of the Rydberg constant.

Equation (9) with V_{eff} from (10) is the hydrogen Schrödinger equation; its eigenvalues are the familiar Rydberg series

$$E_n = -\frac{m_e c^2 \alpha^2}{2n^2} \approx -\frac{13.6\,\text{eV}}{n^2}, \tag{11}$$

with the prefactor expressed entirely in SSV constants $\{m_e, \alpha, c\}$ that Paper I already takes as fixed. The hydrogen-like bound-state spectrum therefore follows from the LogSE low-amplitude long-wavelength limit together with the chiral-shear identification of electromagnetism (Paper I for $\alpha = c_\perp/c$, Paper II [3] for the Coulomb statics), adding no input beyond the ones already conceded. "No fresh input" is the correct claim; "derived" is not, since α enters (10) empirically.

The same construction applies to any heavy SSV source defect. The effective potential is set by the long-wavelength chiral-shear field of that source, and the bound-state spectrum follows by the standard Schrödinger calculation in (9). The LogSE programme does not need to recompute hydrogen; it needs to verify that (10) is the correct linearised field, which is the content of Paper I's electromagnetism sector.

3.2 Quantisation as circulation topology

The phase in (1) must be single-valued around any closed loop that does not cross a zero of Ψ. Therefore

$$\oint_C \nabla S \cdot d\mathbf{l} = 2\pi n\hbar, \qquad n \in \mathbb{Z}. \tag{12}$$

Using (2), this is the circulation condition

$$\oint_C \mathbf{v} \cdot d\mathbf{l} = \frac{2\pi n\hbar}{m}. \tag{13}$$

Quantisation is therefore tied to the topology of the phase field. It is not inserted as a separate rule for allowed orbits; it is the consequence of a single-valued order parameter with vortex cores acting as excluded phase singularities. The detailed spectrum of any bound system depends on its effective potential and boundary conditions, but the origin of discrete winding numbers is hydrodynamic. This is consistent with the topological spin derivation of Paper V [4] and the vortex-defect picture of Paper I [1].

4 Uncertainty as Hydraulic Bound

The standard uncertainty relation follows from Fourier analysis of wave packets:

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}. \tag{14}$$

SSV adds a physical interpretation of the same relation. A perturbation localised to a short length scale must include large gradients of $\sqrt{\rho}$ and phase. Through (6), those gradients carry a quantum-pressure cost $\Delta E_Q \sim \hbar^2/(2m\,\Delta x^2)$. Pushing Δx downward therefore forces a larger spread in momentum modes and a larger hydraulic stiffness penalty, arriving at (14).

This section does not replace the standard proof. The proof remains the wave-packet Fourier theorem. The SSV claim is that the theorem has a medium-level interpretation: localisation is limited because the order parameter cannot be squeezed arbitrarily without exciting the pressure and phase-gradient channels of the fluid. The uncertainty principle is the hydraulic stiffness of the vacuum medium made quantitative.

4.1 Saturation by the Gausson

Rejected-branch record — this section is not current (#189)

This section is retained deliberately, as a record of a route that does not work. It is kept rather than deleted for the same reason Paper I keeps its rejected sign visible: a negative result that is explicit in one place and invisible in another is the defect the #182 audit was created to find. **Nothing below should be read as a current claim of the programme.** Both findings are now machine-checked (`logse_gaussian`; report `logse-gaussian-issue189`).

(i) **The Gausson does not exist on the adopted branch.** Substituting the Gaussian ansatz into the LogSE and requiring the x^2 terms to cancel fixes the width uniquely, and its *sign* is set by the sign of the logarithmic term. In the convention this section prints, $\sigma^2 = +\hbar^2/(2mb)$ and Eq. (16) is correct. In the convention Paper I adopted (#183), $\sigma^2 = -\hbar^2/(2mb) < 0$ and **no normalisable Gaussian exists at all**. Paper I rejected the attractive branch because it makes the uniform vacuum modulationally unstable and cannot support $c_s = c$. On the branch actually adopted, §4.1 has no solution to stand on, and particles must be *topological* rather than bright solitons.

The distinction matters for what is being conceded: this section did not miscalculate. It wrote a correct equation for a theory the series has since rejected.

(ii) **The $\hbar/2$ is imported, not derived — on either branch.** The claim below is that the prefactor comes "directly from the LogSE itself". The offered evidence is that the result is independent of the Gausson width σ, and therefore of b. That independence is the tell: *every* normalised Gaussian saturates $\Delta x\,\Delta p = \hbar/2$, for any width, with b entering nowhere. It is a property of Gaussians, not of the LogSE. The computation performed — Gaussian moments giving $\sigma^2/2$ and $\hbar^2/2\sigma^2$, then multiplying — *is* the standard wave-packet calculation. The LogSE contributed only the claim that its stationary state is Gaussian, and under (i) it has no such state.

That saturation is specific to Gaussianity rather than to normalisability is checked against a control: the normalised state $\psi \propto e^{-|x|/\lambda}$ gives $\Delta x\,\Delta p = (\sqrt{2}/2)\,\hbar > \hbar/2$. Without that control the observation would be empty.

The uncertainty principle may still be recoverable as "hydraulic stiffness of the vacuum" in the sense this paper intends, but not by this route. Whether a *topological* defect can do what the bright soliton cannot is open, and shares a root with #196 and #178: the bare one-component order parameter may not carry the required structure, in which case the replacement is a change of theory rather than a substitution.

The remainder of this section is the rejected-branch material referred to above.

The precise prefactor 1/2 in (14) was taken to follow directly from the LogSE itself, without importing it from the standard wave-packet calculation, by computing the uncertainty product on the LogSE's natural coherent state. The logarithmic Schrödinger equation

$$i\hbar\,\partial_t\Psi = -\frac{\hbar^2}{2m}\,\nabla^2\Psi - b\,\Psi\,\ln\Big(|\Psi|^2/\rho_0\Big) \tag{15}$$

admits an exact Gaussian stationary solution, the *Gausson* of Bialynicki-Birula and Mycielski [5]:

$$\Psi_G(x,t) = \frac{1}{(\pi\sigma^2)^{1/4}}\,\exp\!\left(-\frac{x^2}{2\sigma^2}\right)\,\exp(-i\,E_G\,t/\hbar)\,, \qquad \sigma^2 = \frac{\hbar^2}{2m\,b}, \tag{16}$$

where the Gausson width σ is fixed by the LogSE structure constants m and b. Unlike the linear-Schrödinger Gaussian, the Gausson is a *true soliton*: the Gaussian shape is preserved exactly under LogSE evolution; this is the defining property that singles out the logarithmic nonlinearity [5].

The position and momentum variances on the Gausson are evaluated directly from Gaussian integrals:

$$(\Delta x)^2 = \int x^2\,|\Psi_G|^2\,\mathrm{d}x = \frac{\sigma^2}{2}, \tag{17}$$

$$(\Delta p)^2 = \int \big|-i\hbar\,\partial_x\Psi_G\big|^2\,\mathrm{d}x = \frac{\hbar^2}{2\sigma^2}, \tag{18}$$

where the second line uses $\partial_x\Psi_G = -(x/\sigma^2)\,\Psi_G$ and a single Gaussian integral. The product is

$$\Delta x\,\Delta p\big|_{\text{Gausson}} = \sqrt{\frac{\sigma^2}{2}}\cdot\sqrt{\frac{\hbar^2}{2\sigma^2}} = \frac{\hbar}{2}, \tag{19}$$

which is independent of the Gausson width σ and therefore independent of the LogSE coupling b.

The $\hbar/2$ prefactor is imported, not derived

The $\hbar/2$ in (19) is not earned by the LogSE. Two independent reasons, either sufficient.

Under (i) the LogSE has no normalisable Gaussian on the adopted branch, so there is no coherent state for the bound to be realised on; and under (ii) the width σ cancels out of (19) identically, so (m, b) cannot be fixing anything the answer depends on. The calculation performed in (17)–(19) is the textbook Gaussian wave-packet result, and it returns $\hbar/2$ whether or not the LogSE is anywhere in the problem.

What §4 still supports is the *form* of an uncertainty relation from hydraulic stiffness. The *prefactor* is not earned here. Verdict `MISDERIVED`, recorded in `logse-gaussian-issue189` and guarded against silent re-upgrade by `logse_gaussian`.

5 Measurement as Reconnection

5.1 Irreversibility from topology change

The measurement argument is kept narrow. A measurement is a physical interaction that correlates a microscopic degree of freedom with a macroscopic apparatus by driving some part of the combined medium into an irreversible reconnection or wake-writing event.

Before measurement, a superposition is represented by a coherent field configuration,

$$\Psi = \sum_k c_k \psi_k. \tag{20}$$

As long as the relevant branches remain phase-coherent, interference terms are available. A measurement interaction is the regime in which the probe, apparatus, and surrounding medium cross a threshold where topology changes, phonons are emitted, or persistent wake degrees of freedom are written—the same irreversibility mechanism described for time in Paper III [6]. Reversing the event would require restoring the original phase relation across all of those channels, not merely evolving a small isolated wave packet backward.

5.2 Born weights as a working model

The natural branch weights in (20) are $|c_k|^2$ because the local energy and intensity available to drive an apparatus response are quadratic in the wave amplitude. This gives the usual Born form

$$P_k = |c_k|^2. \tag{21}$$

At the present stage this is a consistency argument, not a full derivation.

Basin-volume derivation. The basin argument is concrete enough to write down. Let the apparatus contain a reconnection-threshold degree of freedom that fires when the local energy density driving it exceeds a fixed value $\varepsilon_\star$, with the rate of firing controlled by the perturbative coupling of the incoming field to the threshold-bound mode. For an incoming field $\Psi = \sum_k c_k\,\psi_k$, the local energy density delivered to the apparatus by branch k is $\varepsilon_k = |c_k|^2\,|\psi_k|^2$. Near the threshold, the firing rate is linear in the available energy density (the leading term in the coupling; higher-order corrections are subleading in $\varepsilon_k/\varepsilon_\star$ near the apparatus operating point):

$$\Gamma_k = \kappa\,|c_k|^2, \qquad \kappa \equiv \int_{\text{apparatus}} g\,|\psi_k|^2\,\mathrm{d}V, \tag{22}$$

where g is the apparatus–field coupling kernel and the apparatus-volume integral is the same for every branch k that uses the same apparatus mode. Over a short observation window τ with $\kappa\,\tau \ll 1$, the probability that branch k has triggered the reconnection is $\Gamma_k\,\tau = \kappa\,\tau\,|c_k|^2$, and the conditional probability that branch k is the one that fires, given that exactly one branch fires in the window, is

$$P_k = \frac{\Gamma_k}{\sum_j \Gamma_j} = \frac{|c_k|^2}{\sum_j |c_j|^2} = |c_k|^2, \tag{23}$$

where the last step uses the standard normalisation $\sum_j |c_j|^2 = 1$.

The basin argument does not derive the Born exponent

This is a consistency argument, not a derivation. The exponent is *assumed, not obtained*. The chain is

$$\varepsilon_k = |c_k|^2|\psi_k|^2 \longrightarrow \Gamma_k \propto \varepsilon_k \longrightarrow P_k \propto |c_k|^2,$$

and the first arrow is the Born weight itself — the premise that the energy density delivered by a branch is *quadratic* in its amplitude. It is the same assumption the preceding paragraph offers as the informal consistency argument. Cancelling κ removes a factor common to all branches; it cannot change an exponent that entered at the first step. This is the error of §4.1 in a second place: what is put in as an amplitude-squared is recovered as an amplitude-squared, and the intervening algebra is where the appearance of derivation comes from.

What the argument *does* establish, conditionally and usefully: given (a) rate linear in delivered energy density, (b) a threshold mode common to every branch, so that κ is genuinely k-independent, and (c) unit normalisation, the reconnection-threshold model reproduces Born weights rather than some competing rule. That is a real constraint on the measurement model. It is not a derivation of the exponent, which is what makes the Born rule hard.

A derivation would have to obtain $\varepsilon_k \propto |c_k|^2$ from the medium dynamics without assuming the quadratic form — or replace the argument with one that fixes the exponent from a symmetry or a counting principle. Neither is in hand.

Notation, recorded with it: κ in (22) is defined by an integral containing ψ_k, so as written it carries a k index that the cancellation requires it not to have. Assumption (b) is what removes it, and it should be stated as an assumption rather than hidden in the notation.

5.3 What this does and does not claim

This picture does not require consciousness, an external classical realm, or a new postulate of collapse. It also does not yet derive a complete theory of detector statistics. Its useful claim is more modest: SSV supplies a candidate physical channel for irreversibility during measurement, namely topology change and wake formation in the medium. This is consistent with, but not identical to, decoherence programmes that source irreversibility from environmental entanglement; the SSV version sources it from physical medium events.

6 Discussion

6.1 Position in the series

Paper VII-a is the QM-limit paper in the SSV series. Its inputs are the Madelung structure of the LogSE (Papers I–IV) and the topological defect picture (Paper I, Paper V). Its output—the Schrödinger limit, circulation quantisation, hydraulic uncertainty, and reconnection measurement—is passed forward as a standing result. Claims about emergent geometry and the Einstein-equation limit are in Paper VII-b; cosmogony is in Paper VIII.

6.2 Relation to the Madelung and pilot-wave literature

The Madelung [7] representation is well-established; this is not a new mathematical result. The SSV contribution is the physical identification of the medium: the Madelung fluid is not an abstract probability fluid but the actual vacuum condensate, with the same medium that supports topological defects, irreversible wakes, and gravitational gradients. The difference from the de Broglie–Bohm pilot-wave interpretation is that SSV does not introduce a guiding wave on top of classical trajectories; the order parameter IS the physical medium in which all dynamics occurs.

6.3 Testable consequences

Two near-term checks follow from the reconnection model.

Q1. Reconnection signatures in controlled measurement systems. If detector clicks are accompanied by irreversible medium events, then sufficiently sensitive optomechanical or superconducting measurement platforms should show excess correlated acoustic, thermal, or phase-noise signatures at the transition from reversible coupling to recorded outcome.

Q2. Born-rule basin scaling. A reduced reconnection-threshold model should test whether branch basin volumes scale as $|c_k|^2$ for simple two-channel apparatus couplings. This is the calculation needed before the measurement section can be promoted from physical interpretation to formal derivation.

6.4 Open problems

1. *Bound-state spectra.* Deriving the hydrogen-like spectrum directly from the LogSE, without treating V_{eff} as an external input, is part of the Paper I programme.

2. *Precise uncertainty prefactor — one route now closed.* The 1/2 in (14) was to emerge from the LogSE coherent-state minimum. That specific route is *closed*, not deferred (#189; §4.1): on the branch Paper I adopted the LogSE has no normalisable Gaussian stationary state, and on either branch the Gaussian uncertainty product is independent of the width and of b, so it cannot carry information about the LogSE. A derivation of the prefactor would have to come from a *topological* minimiser, and no candidate is in hand.

3. *Multi-particle states.* The present treatment addresses single-order-parameter modes. Multi-particle states and exchange statistics require a field-theoretic extension of the Madelung picture, not developed here.

4. *Spin-1/2 and Fermi statistics.* The order parameter Ψ is a scalar; half-integer spin and fermion anti-symmetry require an extension to spinor order parameters or a separate topological argument. This is noted as a gap, not filled here.

7 Conclusion

SSV VII-a isolates the quantum-mechanics limit from the geometry and cosmogony claims elsewhere in the series. The paper's formal core is the Madelung bridge and the Schrödinger limit. Its interpretive core has three parts:

1. Quantisation reflects phase topology: discrete winding numbers follow from the single-valuedness of the order parameter, not from a separate quantisation axiom.

2. Uncertainty reflects hydraulic localisation stiffness: the quantum pressure (6) penalises tight localisation, recovering the Heisenberg bound from the equation of state.

3. Measurement reflects irreversible reconnection or wake-writing in the medium, sourcing irreversibility from topology change rather than from an external classical realm.

That is sufficient to make the paper standalone without requiring it to carry the Einstein-equation or cosmogony arguments.

References

[1] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum — The Geometric Origin of Matter*, 10.5281/zenodo.19651986, 2026.

[2] Norland, S., "Saturated Superfluid Vacuum IV: Gravity as Update-Capacity Gradient," SSV Series Paper IV (2026).

[3] Norland, S., "Saturated Superfluid Vacuum II: Forces as Hydrodynamic Pressure Gradients in the Saturated Superfluid Plenum," SSV Series Paper II (2026).

[4] S. Norland, *Saturated Superfluid Vacuum V: Condensate Black Holes*, SSV programme, 2026.

[5] I. Białynicki-Birula and J. Mycielski, "Nonlinear wave mechanics," *Annals of Physics* **100**, 62–93 (1976).

[6] Norland, S., "Saturated Superfluid Vacuum III: Thermodynamics and Irreversible Time," SSV Series Paper III (2026).

[7] E. Madelung, "Quantentheorie in hydrodynamischer Form," *Z. Phys.* **40**, 322–326 (1927).

A Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (https://github.com/StigNorland/SVT): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

Issues.

- #178 — https://github.com/StigNorland/SVT/issues/178
- #182 — https://github.com/StigNorland/SVT/issues/182
- #183 — https://github.com/StigNorland/SVT/issues/183
- #189 — https://github.com/StigNorland/SVT/issues/189
- #196 — https://github.com/StigNorland/SVT/issues/196
- #205 — https://github.com/StigNorland/SVT/issues/205

Code (each pinned to the commit that last modified it).

- `instruments/paper_vii_a/logse_gaussian.py` @ 9ff05fa — https://github.com/StigNorland/SVT/blob/9ff05fa/instruments/paper_vii_a/logse_gaussian.py

Reports (result notes, pinned to their last commit).

- `papers/SSV-VII-a/results/logse-gaussian-issue189.md` @ 9ff05fa — https://github.com/StigNorland/SVT/blob/9ff05fa/papers/SSV-VII-a/results/logse-gaussian-issue18 md

Change record. This paper states its *current* status only. Corrections, withdrawals and reframings are recorded in https://github.com/StigNorland/SVT/blob/main/papers/SSV-VII-a/CHANGELOG.md, and the full history is in the repository's commits.